

Learning the Causal Algebra of a World, Without Reward

Constructors · Causal DAGs · Active Inference — a method demonstrator and a roadmap toward a self-continual learning agent

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Abstract

We build an agent that is dropped into a world it knows nothing about and figures out the cause-and-effect rules by running its own experiments — with no reward, no goals, and no labels. Its only drive is to reduce uncertainty about how the world works. The system fuses four ideas: causal DAGs (the form of its belief), causal inference (it intervenes, so it learns causation not just correlation), active inference (it picks experiments that teach it the most), and Constructor Theory (it distills what it learns into repeatable, composable "constructors" — transformations it can reliably perform — and grows them into bigger ones). The thesis, demonstrated across eight test worlds and 65 falsifiable checks, is simple: delete the reward term from active inference and the agent still has a complete objective. Capability — reaching goals it was never trained on, and answering "what if I do X?" — falls out of understanding as a by-product. Driven by a self-audit, we then make the agent autonomous: it discovers its own interface (which variables are its knobs), flags a hidden cause vs. noise, does fine continuous control, and runs a self-driven continual loop that detects when the world changes -- both a flipped edge and a newly-appearing gate -- and re-learns with no external schedule. We describe the implementation, the results (65 falsifiable checks), what they tell us, what we deliberately did NOT fake-fix, and the remaining path to a fully self-continual agent.

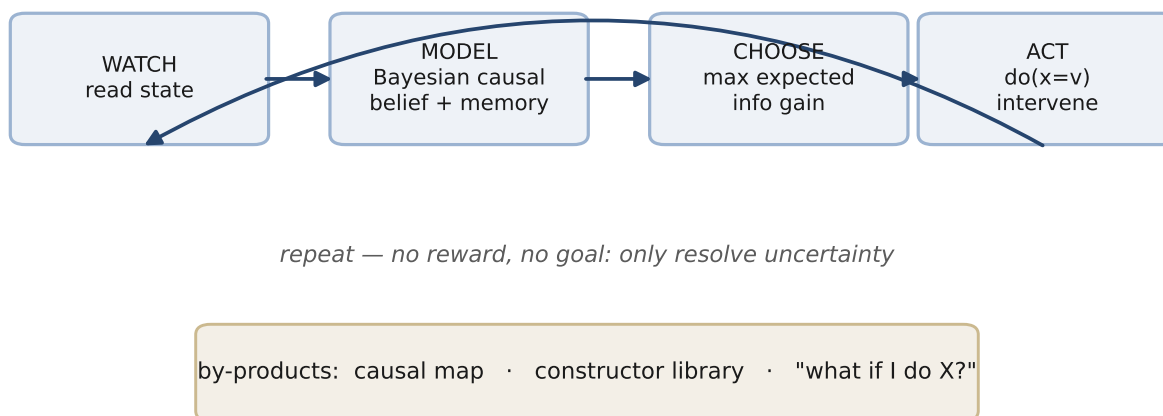


Figure 1. The reward-free perception-action loop. Prediction lives inside the loop as the tool for choosing experiments; the products are the causal map and the constructor library.

1. The idea, in one paragraph

Most learning agents chase a reward. But a child poking at a new toy is not maximising a score — it is

building a model of what the toy does. That is the regime we target. Formally, active inference says an agent should minimise its Expected Free Energy, which splits into two parts:

$$G(\pi) = - (\text{expected info gain}) - (\text{expected reward})$$

Our claim is that the second term is optional. Strike it, and the agent is left with a pure epistemic drive: act so as to learn the most about the world's causal mechanism. Everything else in the system is machinery in service of computing that one quantity well and turning what it learns into things the agent can do.

2. Background, briefly

Causal DAGs and interventions

A structural causal model (SCM) describes a world as variables on a directed acyclic graph, where each variable is a function of its parents plus noise. The crucial operation is the intervention $\text{do}(X=v)$: you reach in and force a variable to a value, severing the arrows that normally set it. Interventions are what separate causation from correlation — two variables can move together because one causes the other, or because a third thing drives both; only forcing one and watching the other tells them apart.

Constructor Theory

Constructor Theory (Deutsch & Marletto) reframes physics around which transformations are possible, rather than around states evolving under laws. A task is a transformation specified by its endpoints (an input region of state space and an output region). A constructor is a thing that, presented with the input, reliably produces the output and is left able to do it again — it is repeatable, like a catalyst. A task is possible if a constructor for it can be built to arbitrarily high reliability. The bridge to causal inference is the heart of this project: an intervention $\text{do}(X=v)$ IS the simplest constructor, and a sequence of interventions is a composite one.

3. Architecture

The system is six cooperating parts. Probabilistic reasoning handles inference and choice; deterministic code handles execution and verification.

- World (world.py) — the hidden ground-truth dynamical SCM. Some variables are actuators the agent may force; the rest are sensors it can only move through the dynamics.
- Model (model.py) — the agent's Bayesian belief about the dynamics; its posterior IS the recovered causal graph, and it yields the information-gain signal in closed form.
- Experimenter (active_inference.py) — reward-free action selection by expected information gain (the epistemic term above).
- Constructors (constructor.py) — Constructor Theory made operational: tasks as boxes, constructors as verified programs, and a composition algebra.
- Synthesiser/Planner (planner.py) — distils constructors from the belief and composes them to reach goals.
- Agent (agent.py) — the loop that ties it together: explore, discover, distil, compose, predict.

4. How it works (implementation)

4.1 The world: a dynamical causal model

State evolves one tick at a time. Forcing an actuator clamps it (overriding its dynamics) and the clamp persists, exactly like dialling a setpoint; sensors then evolve through the graph. This makes time matter: a deep, slow sensor can only be moved by holding upstream knobs over several steps — which is precisely where composing constructors earns its keep.

$$x_{t+1,i} = \sum_j A_{ij} x_{t,j} + b_i + e_{t,i}, \quad e_{t,i} \sim \mathcal{N}(0, \sigma_i^2)$$

4.2 The belief: Bayesian regression whose posterior is the graph

For each sensor we fit one Bayesian linear regression predicting its next value from a feature vector built from the current state. Conjugacy gives a closed-form Gaussian posterior over the weights — not just a best guess but a covariance, i.e. how unsure we still are about each causal link. Two things drop straight out. First, the recovered DAG: an edge $j \rightarrow i$ exists when weight w_{ij} is confidently non-zero ($|\text{mean}|/\text{std}$ beyond a threshold). Interventions are what let the agent drive that std down and zero out spurious links. Second, the experiment value — the expected information gain of observing a transition out of a state with feature vector ϕ :

$$\text{EIG}(\phi) = \frac{1}{2} \sum_i \log(1 + \phi^\top \Sigma_i \phi / \sigma_i^2)$$

This measures shrinkage of uncertainty about the parameters — not raw surprise. That distinction is the whole game: a pure-noise variable is endlessly surprising but teaches nothing once its (parent-free) law is known, so its EIG collapses and the agent stops chasing it. That is the principled cure for the classic "noisy-TV" trap.

4.3 Constructors and composition

A region of state space is an axis-aligned box (intervals per variable), so membership and subset tests are exact and cheap. A constructor bundles a task (precondition box $\rightarrow$ effect box), a program (a schedule of interventions), and a measured reliability. Composition is the engine that grows them:

$$\frac{C_1: P \rightarrow Q \quad C_2: Q' \rightarrow R \quad Q \subseteq Q'}{C_2 \circ C_1: P \rightarrow R}$$

Composability is a causal statement: C_2 may assume its precondition because C_1 guarantees it as a postcondition. Chaining small, individually-verified constructors yields big ones that reach variables no single primitive can move. Reliability is estimated by actually running the program many times in fresh copies of the world — a legitimate repeated experiment — and a task counts as "possible" once reliability clears a threshold over enough trials, the operational stand-in for Constructor Theory's "arbitrarily good approximation".

4.4 Reward-free experiment selection

At each step the experimenter scores candidate interventions by their expected information gain over a short model-rollout and acts on the best (with a little dithering for coverage). When a world has many knobs the joint grid is exponential, so above a cap the experimenter samples candidate settings instead of enumerating them. Two foils — uniform-random and "maximise raw surprise" — are kept to show that the choice of objective, not the machinery, is what matters.

4.5 Distilling and composing constructors

From the learned model the agent mints primitive constructors ("hold this knob here"), discovering each

one's effect by running it. Constructors that move nothing reveal which knobs are causally idle. It then mints conditional (context-dependent) constructors — a knob that is useless from rest but does something once another constructor has set the stage — and iterates, stacking skills bottom-up. To reach a goal it searches over chains of library constructors, verifying promising ones in the real world.

4.6 Two extensions that remove early limitations

- Interaction discovery: rather than being told which products matter, the agent fits a linear model, scans each sensor's residuals for leftover structure, and proposes the product feature (a "gate") that explains it — recovering multiplicative causes it was never handed.
- Nonlinear basis: random Fourier features let the belief fit smooth nonlinear edges (e.g. a saturating tanh), so counterfactual predictions track the curve instead of a straight line.

5. Experiments and results

Eight worlds, each built to exercise one defence, scored by 65 falsifiable checks across seven test suites. Everything below is reward-free.

World	What it tests	Headline result
default	chain + decoy + noise	recover DAG (F1 1.0); reject decoy; ignore noise
hard	weak edge, idle knob	curiosity F1 0.88 vs random 0.84
confounded	hidden common cause	passive infers spurious edge (w 0.85); intervention kills it (0.04)
gated	multiplicative gate	discover gate; compose 2 distinct skills to reach Z
cascade	two-gate cascade	compose 3 distinct skills; reliability 1.0
wide	many distractor knobs	informed planner 13 nodes vs BFS 314
nonlinear	even + saturating edges	linear blind to a_0^2 ; RFF what-if 22x better
default (drifting)	non-stationary world	track flipping edge; spike->settle; prune & rebuild skills

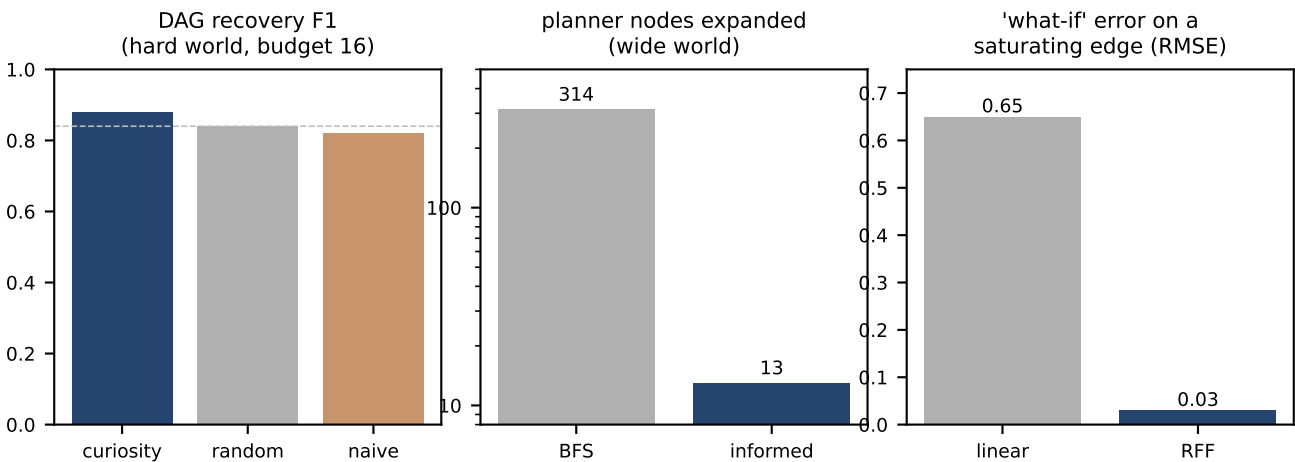


Figure 2. Three headline results, each reward-free: the objective matters under scarcity (left); the informed planner scales (center); a nonlinear basis fixes counterfactuals (right).

5.1 It learns the causal map with no reward

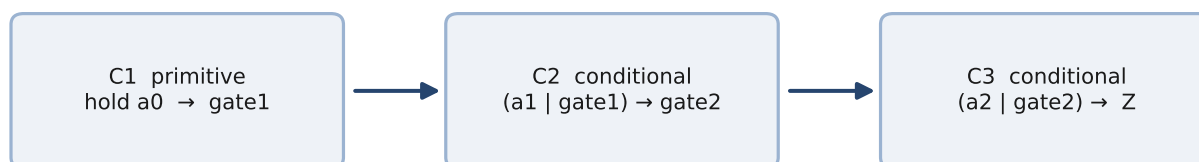
On the default world the agent recovers exactly the true edges ($a_0 \rightarrow \text{chain1}$, $\text{chain1} \rightarrow \text{chain2}$, $\text{chain1} \rightarrow \text{decoy}$) with $F1 = 1.0$. It rejects the decoy — a variable correlated with the target but not its cause — because its interventions break the confound. And it recognises the pure-noise channel as parent-free and stops exciting it, exactly as the information-gain objective predicts.

5.2 The objective matters — but only when it must

On a harder world (a weak edge that must be probed, plus a useless distractor knob and noisy sensors), curiosity beats random $F1$ 0.88 to 0.84 — a modest margin, because the noise-variance recalibration lifted random's floor by suppressing its spurious edges — by concentrating its scarce budget on the informative knob. Honesty cuts both ways: on the easy fully-observed world all objectives tie, and naive surprise only falls into the noisy-TV trap when the agent must also choose **WHAT** to observe — which we show in a dedicated observation-gating test (naive watches noise 83% of the time and learns the real law only 75% of the time; the info-gain agent watches noise 29% and learns it 100%).

5.3 Composition manufactures capability

A deep, slow variable that a single short intervention cannot move (reliability 0.00) is reached by the same primitive chained with itself (reliability ~ 1.0). A multiplicative gate, where a variable rises only when two inputs are both high, is cracked by composing two **DIFFERENT** constructors in order — open the gate, then drive the gated variable. A two-gate cascade needs three distinct context-dependent constructors composed in sequence; the agent discovers and stacks them and reaches the target with reliability 1.0.



composite: $C1 \gg C2 \gg C3$ reaches Z (reliability 1.00)

neither knob alone — and no single knob repeated — can move Z

Figure 3. Constructors compose into bigger constructors. Conditional skills stack bottom-up; the planner chains three distinct ones to crack a two-gate cascade.

5.4 Intervention defeats a hidden confounder

When an unobserved common cause drives two visible variables, a passive observer infers a strong but spurious link between them (weight ~ 0.85). The intervening agent forces one of them, decorrelating it from the hidden cause, and the spurious weight collapses to ~ 0 . This is the textbook reason interventions matter, shown end-to-end.

5.5 The planner scales, and the belief goes nonlinear

In a world padded with useless knobs, uninformed breadth-first search over the constructor library

expands 314 nodes to find the answer; an informed best-first search ordered by the model's predicted distance to the goal expands 13 — the same answer, ~24x cheaper. And two nonlinear edges that defeat a purely linear learner are handled: an even edge (output proportional to a_0 -squared) has zero linear correlation, so a linear model finds no edge at all, yet the agent recovers it as the product $a_0 \cdot a_0$; a saturating tanh edge is predicted with RMSE 0.03 by the random-Fourier model versus 0.65 by the linear one.

5.6 It keeps learning when the world changes

The loop also runs continually. We flip and re-scale a causal edge ($a_0 \rightarrow \text{chain1}$: +0.8, then -0.8, then +1.3) without telling the agent. Three mechanisms keep it current: the belief uses recursive least squares with a forgetting factor, so it tracks recent dynamics rather than averaging over all history; one-step prediction error is monitored, and it spikes the instant the world moves (chain1 error jumps to ~3-4 standard units, then falls back below 0.05 as the belief re-settles); and a consolidation step re-verifies every constructor against the current world, pruning the ones that broke. The recovered weight tracks the true edge across all three regimes; each sign flip prunes the stale skills and the library is rebuilt — the constructor that drove chain1 high by holding $a_0 = +2$ is replaced by one holding $a_0 = -2$, then flipped back. The library stays true to the world, not to its own past.

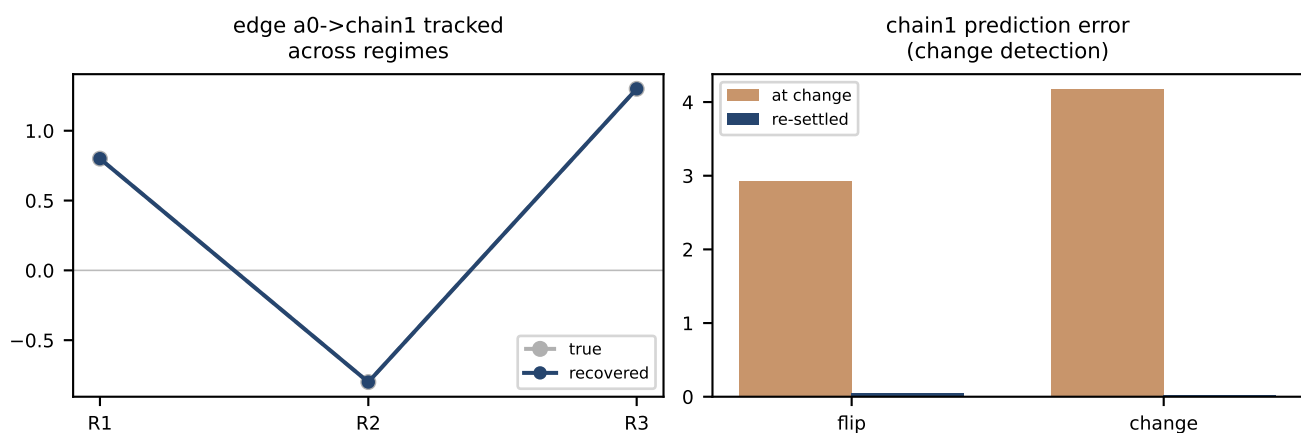


Figure 4. Continual learning. The belief tracks a changing causal edge across three regimes (left); one-step prediction error spikes at each change and re-settles after the agent re-learns (right). Broken skills are pruned and rebuilt each time.

5.7 Closing a self-audit: discovering the interface and running itself

We asked where the system was failing and fixed four gaps, all reward-free. (i) DISCOVER THE INTERFACE: rather than being told which variables are knobs, the agent pokes each one to an out-of-range value and keeps those that hold there (controllability); correct on every world. (ii) HIDDEN CAUSE vs NOISE: a slow unobserved common cause is absorbed into an inflated self-loop, so residual autocorrelation does NOT find it (we checked) -- but its fingerprint is being poorly predicted yet strongly autoregressive, which flags the confounded variable while leaving a pure-noise channel alone. (iii) CONTINUOUS CONTROL: when the coarse +/-2 library overshoots a narrow target, the planner solves for an intermediate setpoint that lands centred in it. (iv) AUTONOMOUS LOOP: the agent watches its own surprise -- standardised by each sensor's noise, so an irreducibly noisy channel cannot raise a false alarm -- and decides for itself when to re-learn. With no external schedule it catches a parametric flip (~56 sigma, prunes the broken skills, relearns the edge) and a structural change, a gate that newly appears (~27 sigma, re-

discovers the interaction), and stays quiet when nothing changed.

We were equally careful about what NOT to claim. Two suspected failures, tested, did not actually bite (an inadmissible-heuristic blow-up on the gated world; false-positive interactions under multiple testing). And several limits are genuine, not fixed: certifying a hidden variable (an AR cause is indistinguishable from a self-loop in one-step data), verification that still assumes a cloneable/resettable world, and untested scale.

6. What this tells us

- Reward is not necessary for competence. A complete, well-defined drive — reduce uncertainty about the causal mechanism — produces an agent that can later be handed arbitrary goals and meet them. Understanding first; capability falls out.
- Curiosity must be about learning, not surprise. Information gain over parameters, not raw predictive surprise, is what avoids the noisy-TV trap and what makes scarce experiments count.
- Causal structure needs intervention. Decoys and hidden confounders both fool passive observation; forcing variables is what makes the recovered graph trustworthy.
- Constructor composition is a genuine engine of capability. Stacking verified transformations produces abilities none of the parts had alone — the operational meaning of "combine sequences of constructors into bigger constructors".
- Honesty about regimes. The objective's advantage, the planner's advantage, and the nonlinear basis's advantage each appear only under the conditions that call for them; we measure where they help rather than asserting they always do.

7. Toward a self-continual learning agent

Section 5.6 shows the loop already surviving a changing world. A fully self-continual agent would keep it turning forever, accumulating and consolidating skills while discovering its own interface and structure. The pieces below are the concrete next steps; the first four are near-term and build directly on what exists.

7.1 Discover the interface (done; certifying the latent is still open)

This is now implemented (5.7): the agent discovers its actuators by controllability probing, and flags hidden state by the poorly-predicted-yet-autoregressive fingerprint. What remains is CERTIFICATION — from one-step observational data a hidden AR(1) cause is indistinguishable from a real self-loop, so the agent can say 'something unobserved drives this' but not identify the latent. That needs interventions on the affected variable (unavailable when it is only a sensor) or an explicit latent-variable model fit by EM / variational inference — the machinery the sibling pi_ct_nsdm project already uses.

7.2 A posterior over structure, not a point estimate

Our per-edge significance test is a thresholded point estimate. The principled object is a posterior over graphs, and the principled acquisition is the mutual information between the next experiment and that posterior (BALD). An ensemble of structures with Bernoulli edge probabilities makes this tractable and turns experiment selection into honest D-optimal design — the same machinery the sibling pi_ct_nsdm project already prototypes.

7.3 An informed, learned planner

The best-first heuristic is greedy and not admissible. Two upgrades: learn a value function over the constructor library (how close does owning skill S leave me to a family of goals?) to guide search, and cache successful chains as new named constructors so the library deepens rather than re-deriving. This is the options framework of hierarchical RL, but with verified, reusable, reward-free skills.

7.4 Off-manifold-robust nonlinear discovery

Random Fourier features predict well on the training manifold but extrapolate poorly off it, and structure is still read from the linear/product blocks. The fix is interventional group-relevance: force a candidate parent across its range and test whether a whole nonlinear feature block for it carries causal weight — a nonlinear analogue of the linear edge test — with Gaussian-process or deep-kernel beliefs for calibrated off-manifold uncertainty.

7.5 Non-stationarity and consolidation (mostly done)

Sections 5.6-5.7 implement this: forgetting, AUTONOMOUS change detection by standardised surprise, library consolidation, and structural re-discovery when a gate appears. What remains turns it from "survives scripted changes" into "survives an open-ended life":

- Skill merging and abstraction: consolidation currently prunes; it should also merge near-duplicate skills and promote frequently-used chains to named macro-constructors.
- Competence progress as a second drive: alongside information gain, reward learnable progress (am I getting better at predicting / achieving?) to allocate effort and to avoid the "dark-room" failure where an agent minimises surprise by doing nothing.
- Empowerment: prefer states from which many transformations remain possible, a constructor-theoretic objective that keeps options open without any external reward.

7.6 The continual loop

Putting it together, the agent we are building toward runs one loop indefinitely: explore the most uncertain frontier; discover new variables, edges, and interactions; distil and verify constructors; compose them into deeper skills; consolidate the library; and detect when the world has changed and the cycle must reopen. No reward is defined at any point. Goals, when they arrive, are solved by composing what the agent already understands — which is, we argue, the right shape for an agent that must keep learning in a world it was never told the rules to.

8. Reproducibility

The system is ~3.4k lines of numpy-only Python. Every claim in Section 5 is asserted by an executable check that exits non-zero on failure.

```
# from the ChemicalWorld directory (project .venv; numpy only)
python -m constructor_causal.demo          # basic: DAG -> library -> compose
python -m constructor_causal.demo_advanced # confounder, gate, observation-gating
python -m constructor_causal.demo_frontier  # interaction discovery + deep cascade
python -m constructor_causal.demo_frontier2 # informed planner + nonlinear
python -m constructor_causal.demo_continual # a world that changes underneath the agent
python -m constructor_causal.demo_autonomous # discovers interface; self-driven loop
python -m constructor_causal.selftest{,_advanced,_frontier,_frontier2,_continual,_autonomous} #
65 checks
```


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